ARY PRASETYO CAHYONO

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A final-year Instrumentation Physics student awaiting graduation, highly interested in Instrumentation Physics, highly interested in working with Internet of Things (IoT), artificial intelligence (AI), and hardware integration. Experienced in microcontroller programming, system automation, and mentoring students in basic instrumentation.

Work Experience

Lembaga Minyak dan Gas Bumi (LEMIGAS) - South Jakarta, Jakarta Apr 2024 - Aug 2024
Project Management & IoT Intern

- Collaborated with a 13-person team to design and build an IoT-automated hydroponics system from scratch.
- Involved in physical hardware assembly such as producing a panel box for monitoring hydroponics system that consists of ESP32 microcontroller and 5 different sensors.
- Handled technical coding such as sketch using Arduino IDE and other firmware that runs the ESP32 microcontroller and its sensors for hydroponics system and MQTT network.
- Established MQTT connection between the instrument and the server for transmitting and receiving data on both sites with 9 topics.
- Designed and developed a Graphical User Interface (GUI) dashboard using node-red that displays 5 different data for real-time system monitoring.

Smart Urban Farming (SUF) - South Tangerang, Banten Jun 2024 - Sep 2024
Electrical Division Member

- Assisted in assembling, wiring, and troubleshooting plug-and-play IoT monitoring kits for urban agriculture.
- Collaborated with the electrical team to implement wiring diagrams and integrate microcontrollers with various environmental sensors.
- Performed technical maintenance and sensor calibration to ensure the stability of the automated farming systems.

Education Level

Universitas Islam Syarif Hidayatullah Jakarta - South Tangerang City, Banten Sept 2021 - May 2026
Undergraduate in Physics, 3.57/4.00

- Appointed as basic electronics lab assistant, impacting 50+ physics students major
- Appointed as advanced electronics lab assistant, impacting 30+ physics students major
- Appointed as jury prototype competition at study orientation period, impacting 10+ teams
- Appointed as robotics speaker at orphanage, impacting 20+ attendee

Organizational Experience

HIMAFIWEEK - South Tangerang City, Banten Jun 2023 - Oct 2023
Staff of Goods and Facilities Procurement Division
- Provided necessary items during the activity or contest for 10 matches.
- Procured facilities for competition activities such as making overlay for online competition or rented sports facilities for 10 matches.

Robotics Club of HIMAFI - South Tangerang City, Banten Mar 2024 - Jan 2025
Secretary
- Managed all organizational documentation, including meeting minutes, correspondence, and membership databases, ensuring a streamlined administrative workflow.
- Coordinated the scheduling of weekly club activities and technical workshops for over 20+ members of the Physics student’s major.
- Organized the club's undergraduate mentoring initiative, impacting 20+ physics students major with robotics interest.
- Mentored club's undergraduate mentoring initiative, impacting 20+ physics students major.
- Assisted the Chairman in overseeing the club's administrative operations and ensuring all activities aligned with the organization's objectives for the 2025 period.
- Aimed goals during the period by making project for orphanage as education tools.
- Supervised the internal communication channels to ensure effective dissemination of information regarding club events and project deadlines.

EKSPEDISI (Physics Freshmen Orientation) - South Tangerang City, Banten Oct 2022
Head of Security, Health, and Cleanliness (K3) Division

- Led the K3 division in overseeing the security operations, medical response, and sanitation protocols during the 3-day physics freshmen orientation program from October 28 to 30, 2022.
- Managed crowd control, enforced event regulations, and provided immediate first-aid assistance to ensure the safety and well-being of all participants.
- Coordinated waste management and maintained strict venue hygiene standards throughout the event, fostering a safe, clean, and conducive environment for new students.

Skills, Achievements & Other Experience

Soft Skills*:* Leadership & Team Management, Technical Mentoring & Communication, Problem Solving, Hardware/Software Troubleshooting, Project Management, Risk Management & Event Safety, Cross-functional Collaboration.

Hard Skills

- **IoT & Automation**: Node-RED, System Automation, Graphical User Interface (GUI) Development.
- **Microcontrollers & Electronics:** Wemos D1 R1, Arduino, Sensor Calibration, Instrumentation Physics, PCB/Circuit Assembly, Advanced Instrumentation (Scintillation Detectors, Thermopiles, Tesla meters), Statistical Data Processing.
- **Programming & AI:** C/C++, OpenAI Whisper Integration, API Implementation, Data Analysis (Linear Regression), Cassy Lab Software.
- **IT & Hardware:** PC Hardware Building, Performance Optimization, IT Troubleshooting.
- **Languages:** English (Professional Working Proficiency), Arabic (Elementary Proficiency).

Projects

- **Voice-Controlled Robotic Hand with AI Integration (Final Thesis)** (2025): Developed an advanced assistive robotic hand system utilizing the OpenAI Whisper model for high-accuracy speech recognition. Engineered the integration between the AI model and a Wemos D1 R1 microcontroller to translate vocal commands into precise mechanical finger movements and palm actuations.
- **Joystick-Controlled Robotic Arm Prototype** (2025): Engineered and assembled a miniature robotic arm utilizing an Arduino microcontroller and multiple servo motors, implementing precision manual control through a dual-axis joystick interface.
- **Automated Hydroponics Monitoring System** (2024): Designed and implemented a real-time monitoring system for hydroponic plants, utilizing a custom Node-RED dashboard for automated control and environmental data visualization.
- **IoT Automatic Parking Barrier System** (2024): Developed an automated parking detection and access system by integrating ultrasonic sensors and NFC modules, utilizing the ThingSpeak platform for remote IoT data transmission and monitoring.
- **Arduino Real-Time Monitoring GUI** (2023): Built a hardware-software integrated system utilizing an Arduino R3, an LDR sensor, and a joystick. Programmed a custom graphical user interface (GUI) to display real-time dynamic graphs of sensor data and joystick coordinates.

Advanced Laboratory Research

- **Atomic Physics (Franck-Hertz)** (2023): Analyzed the excitation energy of Mercury (Hg) atoms through precise voltage and current measurements, calculating standard deviations to verify experimental accuracy against literature values.
- **Electromagnetic & Material Instrumentation** (2022 - 2023): Conducted Hall Effect and Specific Electron Charge experiments. Utilized Teslameters, tangential B-probes, and Helmholtz coils to measure magnetic flux, Hall voltage, and Lorentz force interactions. Applied linear regression methods for precise data calculation and error analysis.
- **Radiation & Spectroscopy Analysis** (2022)**:** Operated NaI (TI) Scintillation detectors, Geiger-Muller counters, and MCA Boxes utilizing Cassy Lab software to calibrate gamma energy spectrums and measure radiation absorption coefficients across various material mediums.
- **Thermal & Wave Sensors** (2022): Utilized Moll's thermopiles, NiCr-Ni temperature sensors, and Gunn oscillators to measure black body radiation intensity and map microwave electric field distributions.